

An Early Inverse Bridge for Affine-Transfer Induction

Collatz proof project — September 5, 2026

Write $\mathcal{R}(n)$ for reaching one under the shortcut Collatz map T , and $\mathcal{A}(u) = [\mathcal{R}(3u + 2) \Rightarrow \mathcal{R}(27u + 20)]$. The preceding cycle-based construction needed convergence of a base target. Here an immediate inverse step yields a conditional induction certificate without that premise. The universal transfer statement remains unproved.

The early predecessor. For every natural w , put $u = 3w + 2$ and $v = 2w + 1$. Then

$$v < u, \quad T(3v + 2) = 3u + 2, \quad T(27v + 20) = 81w + 71,$$

whereas $27u + 20 = 81w + 74$. Thus $\mathcal{R}(3u + 2)$ and the induction hypothesis $\mathcal{A}(v)$ supply convergence of $81w + 71$. A suitable orbit merge with $81w + 74$ completes transfer at u . The source predecessor is available before inspecting any target trajectory.

Theorem 1 (Uniform early inverse family). *For every natural Q , define*

$$u = 59 + 192Q, \quad v = 39 + 128Q.$$

Then $v < u$ and

$$T(3v + 2) = 3u + 2, \\ T^7(27v + 20) = T^6(27u + 20) = 227 + 729Q.$$

Consequently $\mathcal{A}(v) \Rightarrow \mathcal{A}(u)$.

Proof. The first identity follows by applying the odd shortcut step to $3v + 2 = 119 + 384Q$. For the second, the base paths satisfy

$$T^7(1073) = 227, \quad j_7(1073) = 3, \quad T^6(1613) = 227, \quad j_6(1613) = 2.$$

The input increments are $2^7 \cdot 27Q$ and $2^6 \cdot 81Q$, so binary-cylinder transport gives identical endpoint increments $729Q$. Also $u - v = 20 + 64Q > 0$. Given $\mathcal{R}(3u + 2)$, transport convergence backward one step, apply $\mathcal{A}(v)$, and then use the common endpoint to conclude $\mathcal{R}(27u + 20)$. Lean checks the identities, inequality, and conditional implication for every Q . $\square$

Exact search criterion. For a binary cylinder $w = r + 2^tQ$, let x, y be the t -step endpoints of $81r + 71$ and $81r + 74$, with odd counts j, k . The endpoint increments are $3^{j+4}Q$ and $3^{k+4}Q$. Hence $x = y$ and $j = k$ give a uniform merge. The recursive parameters are

$$u = 3r + 2 + 3 \cdot 2^tQ, \quad v = 2r + 1 + 2^{t+1}Q < u.$$

These are early certificates: the criterion checks exact finite orbit identities, not eventual target cycle entry.

Finite evidence. At depth 12 the search finds 104 first rules. They cover 407 of the 4096 binary residues of w , leaving 3689 unresolved. This is coverage inside $u \equiv 2 \pmod{3}$, not of all parameters. Three tests compare complete small-depth searches with direct orbits, replay all rules on three quotient lifts, and check the first rule $r = 19, t = 6$. The census is Python evidence; the displayed family is kernel checked.

Remaining obligation and verification. This family supplies a genuine smaller-parameter induction step without an unproved target-convergence premise. It does not establish universal rule coverage. In particular, parameters $2^k - 1$ lie in the other two classes modulo three, so this first inverse template does not resolve the preceding bootstrap obstruction. Other templates or a global argument are still needed. The three public declarations in `Collatz.EarlyInverseBridge` enter the axiom audit. The search is `analysis/early_inverse_bridge_search.py`. No mathematical priority claim or proof of Collatz is asserted.